

Regression Analysis Report

Introduction

Dataset Overview

This report analyzes the PakWheels Used Car Dataset containing 77,878 car listings from the Pakistani automotive market spanning 1990-2022. The dataset includes car specifications, pricing, and market characteristics for regression-based price prediction.

Objectives

Build a regression model to predict used car prices

Identify key factors influencing car pricing in Pakistani market

Provide insights for buyers, sellers, and market analysts

Data Cleaning and Preparation

Missing Value Handling

Assembly Column: Dropped due to 68% missing values

Numerical Features: Filled with median (year, mileage, engine)

Categorical Features: Filled with mode (transmission, fuel, body, make, city, model, color)

Smart Categorical Encoding Strategy

Three-tier approach based on cardinality:

Low Cardinality (≤ 5 values): One-Hot Encoding $\rightarrow$ transmission, fuel

Medium Cardinality (5-130 values): Label Encoding $\rightarrow$ body, make, registered

High Cardinality (> 130 values): Target Encoding $\rightarrow$ city, model, color

Key Benefits:

Avoided dimensional explosion (prevented 1,340+ columns)

Target encoding captured price relationships for high-cardinality features

Preserved interpretability with mapping files

Feature Standardization & Data Split

Standardization: Applied to original numerical features only

Train-Test Split: 80-20 ratio (62,302 train / 15,576 test samples)

Final Features: 13 after encoding

Data Analysis and Visualization

Target Variable (Price) Analysis

Distribution Characteristics:

Extreme right-skew with long tail

Market Segments: Budget (3.1%), Mid-range (22.6%), Premium (74.3%)

Key Finding: Log transformation creates near-normal distribution

Outliers: 880 cars above ₹2.1 crores

Numerical Features Insights

Feature Correlations with Price:

Engine: +0.45 (strongest numerical predictor)

Year: +0.27 (newer cars cost more)

Mileage: -0.18 (higher mileage reduces price)

Key Patterns:

Year: Most cars are recent (2018-2020), exponential price growth post-2015

Mileage: Extreme right-skew, most cars <100k km

Engine: Popular sizes at 1000cc, 1300cc, 1600cc

Categorical Features Impact

Market Segmentation:

Fuel Type: Petrol > Hybrid > Diesel (pricing hierarchy)

Body Type: SUV/Sedan > Hatchback (clear premium vs economy)

Transmission: Manual $\approx$ Automatic (similar median prices)

Brands: Luxury brands (Mercedes/BMW) show extreme outliers

Feature Importance Ranking

Top Predictors (by correlation):

Model (target-encoded): 0.63 - Most critical predictor

Color (target-encoded): 0.51 - Surprisingly important

Engine: 0.42 - Key numerical feature

Transmission: 0.35 - Moderate importance

Year: 0.28 - Age factor

Critical Finding: Categorical features significantly outperform numerical features for price prediction.

Data Quality Assessment

Strengths:

Large sample size (77,878 cars) provides statistical power

Comprehensive feature mix (technical + market characteristics)

32-year temporal coverage

Limitations:

Premium market bias (74% cars >₹15L)

Limited budget car representation (3.1% <₹5L)

880 extreme outliers may impact modeling

Key Findings & Recommendations

Market Insights

Premium-focused dataset primarily represents higher-end vehicles

Model-specific pricing matters more than brand reputation

Target encoding strategy highly effective for categorical features

Engine capacity is the most predictive numerical feature

Modeling Recommendations

Use log(price) as target variable for better distribution

Prioritize categorical features in model selection

Consider outlier removal for cars >₹2.1 crores

Expect best performance on premium cars (₹15L+ segment)

Technical Success

Smart encoding prevented dimensional explosion while preserving information

Target encoding captured complex categorical-price relationships

Robust preprocessing pipeline with 99%+ data retention